Personalized Movie Recommendation System

Final Report

Recommender Systems

Group Project

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1. Introduction

Recommendation systems help users discover relevant items by learning from user preferences and item characteristics. The aim of this project was to develop a personalized movie recommendation system using the MovieLens ml-latest-small dataset. Four recommendation approaches were implemented and compared: a popularity-based baseline, a content-based recommender, collaborative filtering, and a hybrid recommender. The primary objective was to determine whether personalized recommendation techniques outperform recommending the same popular movies to every user.

2. Dataset and Data Preprocessing

The MovieLens ml-latest-small dataset consists of user ratings, movie metadata and user-generated tags. The project uses ratings.csv to capture user-item interactions, movies.csv for movie titles and genres, and tags.csv to enrich movie descriptions. The datasets were merged using movie identifiers, and genres and tags were combined before applying TF-IDF vectorization. To simulate a real-world recommendation scenario, an 80/20 temporal train-test split was used. Older ratings were used for training while newer ratings were reserved for testing, preventing data leakage.

3. Methodology

Four recommendation models were implemented. First, a popularity-based recommender ranked movies using Bayesian weighted ratings, providing a simple benchmark. Second, the content-based

recommender represented movies as TF-IDF vectors generated from genres and tags. A rating-weighted user profile vector was constructed from highly rated movies, and cosine similarity was used to rank unseen movies. Third, collaborative filtering was implemented using Item-Item Collaborative Filtering and Matrix Factorization (SVD). Item-Item Collaborative Filtering recommends movies with similar historical rating patterns, while SVD learns latent user and item factors from the sparse rating matrix. Finally, a hybrid recommender combined content-based and collaborative scores using a weighted parameter (α), balancing semantic similarity with collaborative user behaviour.

4. Evaluation and Results

The recommender system was evaluated using Top-N recommendation metrics: Precision@10, Recall@10, HitRate@10 and NDCG@10. These metrics measure recommendation relevance, coverage and ranking quality. The hybrid recommender achieved the best overall performance, demonstrating that combining content information with collaborative filtering produces more accurate and personalized recommendations than either approach individually. A Streamlit application was also developed to demonstrate the recommender interactively, allowing users to compare recommendation algorithms, generate Top-10 recommendations and view explanations for suggested movies.

5. Limitations, Future Work and Conclusion

The project has several limitations, including the cold-start problem for new users and movies, sparse rating data, limited movie metadata and the lack of contextual information such as mood or time of day. Future work includes replacing TF-IDF with semantic embeddings such as SBERT, exploring Graph Neural Networks and neural recommender architectures, and incorporating context-aware recommendation techniques. In conclusion, the project demonstrates that personalized recommendation methods outperform popularity-based recommendations. The hybrid recommender provides the best balance between personalization, recommendation accuracy and explainability, highlighting the benefits of combining multiple recommendation strategies.

References

1. MovieLens Dataset (GroupLens Research).
2. Streamlit Documentation.